

Portfolio Summary

By: Osemudia Ogedengbe

Class: ITAI-2373-Natural Language Processing

Github Portfolio:

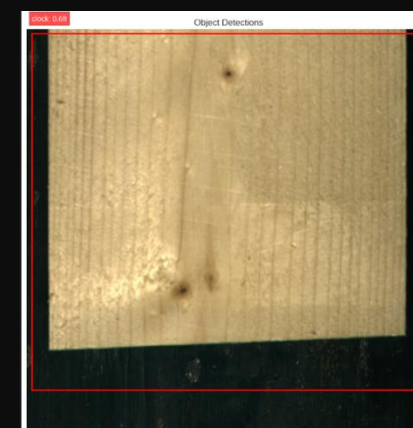
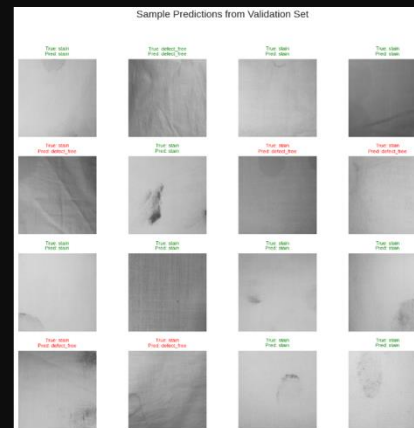
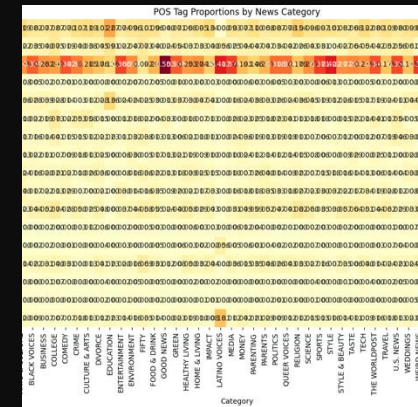
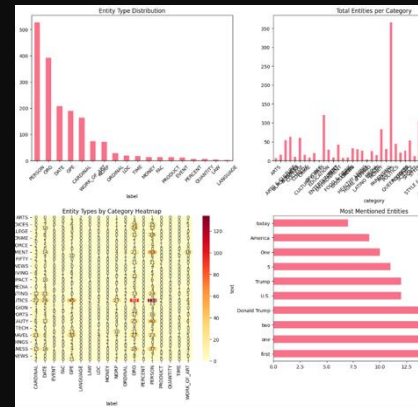
<https://github.com/ItsAlright/Osemudia-Ogedengbe-HCC-AI/tree/main>

Learning Journey

- For this course I've gone through a series of labs that taught me concepts regarding Natural Language Processing. Notable labs during this journey include:
 - Lab 02 – Where I was taught NLP Processing Techniques
 - Lab 04 – Where I learned more about how Models interpret and handle text.
 - Lab 05 – Where I learned more about Part-of-Speech Tagging
- Over the weeks that I learned more about NLP, I became more invested in it than I would've before.

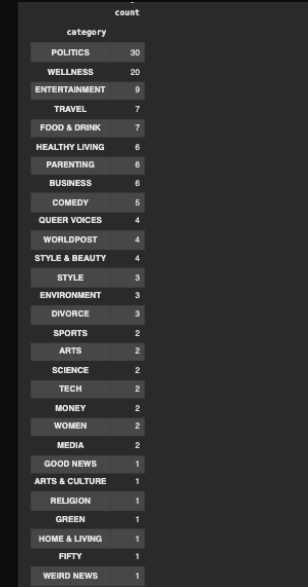
Featured Projects

- Two of the projects I was working on that I'm rather proud of include:
- News Bot – A news analysis platform that is capable of analyzing news articles with advanced NLP techniques.
- Sofa Manufacturer – Despite its name, it's a computer vision model that identifies the qualities of the materials used for the fabric, and frame of the sofa.

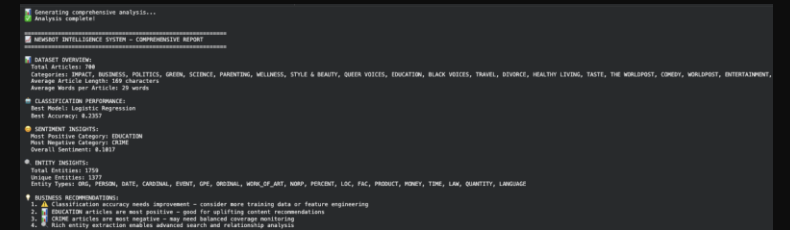


Skills & Competencies

- These are two screenshots of two versions of a News Categorization Bot. Though with the second version I had some assistance I learned a lot when making both.
- Most notably the preparations for situations that won't go as intended. Like how I made two sub-versions of the first version just in case I failed to get one out.



category	count
POLITICS	30
WELLNESS	20
ENTERTAINMENT	9
TRAVEL	7
FOOD & DRINK	7
HEALTHY LIVING	6
PARENTING	6
BUSINESS	6
COMEDY	5
QUEER VOICES	4
WORLDPOST	4
STYLE & BEAUTY	4
STYLE	3
ENVIRONMENT	3
DIVORCE	3
SPORTS	2
ARTS	2
SCIENCE	2
TECH	2
MONEY	2
WOMEN	2
MEDIA	2
GOOD NEWS	1
ARTS & CULTURE	1
RELIGION	1
GREEN	1
HOME & LIVING	1
FIFTY	1
WEIRD NEWS	1



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Generating comprehensive analysis...
Analysis completed.

NEWSBOT INTELLIGENCE SYSTEM - COMPREHENSIVE REPORT

DATASET OVERVIEW:
Total Articles: 100
Categories: IMPACT, BUSINESS, POLITICS, GREEN, SCIENCE, PARENTING, WELLNESS, STYLE & BEAUTY, QUEER VOICES, EDUCATION, BLACK VOICES, TRAVEL, DIVORCE, HEALTHY LIVING, TASTE, THE WORLDPOST, COMEDY, WORLDPOST, ENTERTAINMENT.
Average Article Length: 128 characters
Average Words per Article: 20 words

CLASSIFICATION PERFORMANCE:
Best Model: Logistic Regression
Best Accuracy: 0.920

CONTAINER METRICS:
Most Positive Category: EDUCATION
Most Negative Category: COMEDY
Overall Sentiment: 0.680

INTELLIGENCE SUMMARY:
Total Articles: 100
Total Sentiment: 0.680
Entity Types: ONE, PERSON, DATE, CATEGORY, EVENT, GPE, ORGANIZATION, WORK_OF_ART, MISC, PERCENT, LOC, FAC, PRODUCT, MONEY, TIME, LAW, QUANTITY, LANGUAGE

BUSINESS RECOMMENDATIONS:
1. Classification accuracy needs improvement - consider more training data or feature engineering
2. Classification articles are most positive - good for uplifting content recommendations
3. COMEDY articles are most negative - keep these behind content moderation
4. Each entity extraction module shows accurate search and relationship analysis
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Reflection & Next Steps

- Overall, I'd say that I had a meaningful learning experience throughout this course. And I really liked all the projects that I got to do.
- As for the future though, I do plan on revisiting and upgrading the computer vision model that identifies the quality of sofa components.
- However, I'm still up for other future projects.
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